A13483W1

# SECOND PUBLIC EXAMINATION

**Honour School of Physics Part B: 3 and 4 Year Courses**

**Honour School of Physics and Philosophy Part B**

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## B4: SUB-ATOMIC PHYSICS

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**TRINITY TERM 2018**

**Thursday, 7 June, 2.30 pm – 4.30 pm**

*Answer* **two** *questions.*

*Start the answer to each question in a* **fresh book.**

*A list of physical constants and conversion factors accompanies this paper.*

*The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.*

**Do NOT turn over until told that you may do so.**

**1.** (a) Explain why stable nuclei which are light have atomic numbers approximately equal to half of their mass numbers, whereas for heavier nuclei this is not the case. Hence explain why the decay products of nuclear fission reactions generate antineutrinos. [7]

(b) Fission reactor antineutrinos may be detected using stationary proton targets through the inverse beta decay process $\bar{\nu} + p \to e^+ + n$. Draw a quark-level Feynman diagram for this process, and explain briefly why the cross-section increases with increasing antineutrino energy $E_{\bar{\nu}}$ for antineutrinos with reactor energies. [5]

It is proposed to observe antineutrinos from a compact ${}^{235}_{92}\mathrm{U}$ fission reactor which generates thermal power of 100 MW. The $\bar{\nu}$ detector material is a $\mathrm{C_9H_{10}}$ polymer scintillator.

The table shows, as a function of the $\bar{\nu}$ energy $E_{\bar{\nu}}$, the inverse beta decay cross-section $\sigma_{\bar{\nu}}$ , the number $\frac{\mathrm{d}N_{\bar{\nu}}}{\mathrm{d}E_{\bar{\nu}}}$ of antineutrinos produced per ${}^{235}\mathrm{U}$ fission per unit $\bar{\nu}$ energy, and their product, expressed in consistent units.

| $E_{\bar{\nu}}$ [MeV] | $\sigma_{\bar{\nu}}$ [$10^{-42}$ cm$^2$] | $\frac{\mathrm{d}N_{\bar{\nu}}}{\mathrm{d}E_{\bar{\nu}}}$ [fission$^{-1}$ MeV$^{-1}$] | $\sigma_{\bar{\nu}}\frac{\mathrm{d}N_{\bar{\nu}}}{\mathrm{d}E_{\bar{\nu}}}$ |
|---|---|---|---|
| 2 | 0.03 | 1.48 | 0.04 |
| 3 | 0.26 | 0.88 | 0.23 |
| 4 | 0.67 | 0.45 | 0.30 |
| 5 | 1.27 | 0.18 | 0.23 |
| 6 | 2.03 | 0.02 | 0.04 |

(c) Explain why the rate of observed neutrino interactions is proportional to the integral

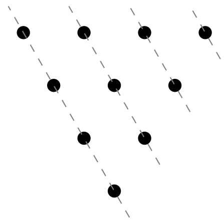
$$I = \int \sigma_{\bar{\nu}} \frac{\mathrm{d}N_{\bar{\nu}}}{\mathrm{d}E_{\bar{\nu}}} \,\mathrm{d}E_{\bar{\nu}}\,.$$

[2]

(d) Given that the detector is placed 7 m from the reactor, find: (i) the rate of fission events; (ii) the value of the integral $I$; (iii) the mass of scintillator required in order that 90 antineutrinos be detected each hour. [11]

[*You may assume that the average energy released per fission of* ${}^{235}\mathrm{U}$ *is* 210 MeV.]

**2.** (a) The $J^P = \frac{3}{2}^+$ decuplet of hadrons is represented below. The dashes indicate lines of equal charge.



| Label | Mass (MeV/$c^2$) | Strangeness |
|---|---|---|
| $\Delta$ | 1238 | 0 |
| $\Sigma^*$ | 1385 | $-1$ |
| $\Xi^*$ | 1532 | $-2$ |
| $\Omega^-$ | ? | $-3$ |

Explain how the spin, parity, strangeness and charges of these hadrons can be explained in the quark model. [9]

(b) The $\Omega^-$ was discovered through the process

$$K^- + p \to \Omega^- + K^0 + X$$

when a $K^-$ beam was directed at a hydrogen bubble chamber target. Assuming the production to have proceeded via the strong interaction, state the quark content of the unspecified $X$ hadron, and draw a quark-flow diagram for this process. [4]

(c) By considering the mass differences between the other states in the decuplet, predict the mass of the $\Omega^-$. Calculate the minimum $K^-$ momentum required for this production reaction, expressing your answer in units of MeV/$c$. [6]

(d) The $\Omega^-$ particle was observed to decay through the following processes

$$\Omega^- \to \Xi^0 + \pi^- \, .$$

Draw a Feynman diagram illustrating this decay. [2]

(e) The momentum of the $\Omega^-$ was calculated to be 2015 MeV/c, and it travelled a distance of 2.5 cm. Estimate the lifetime of the $\Omega^-$ in seconds. [4]

**3.** (a) What is the value of the *baryon number* $\mathbb{B}$ for the fundamental fermions and anti-fermions of the Standard Model, and of the baryons and mesons? [5]

(b) By considering charge $\mathbb{Q}$, baryon number, and lepton number $\mathbb{L}$, explain which of the following interactions are permitted in the Standard Model:

(i) $p + p \to n + p + \pi^+$

(ii) $p + e^- \to \rho^0$

(iii) $\mu^- \to e^- + \nu_\mu + \bar{\nu}_e$

(iv) $\nu_e + e^+ \to d + \bar{u}$.

Draw a Feynman diagram or quark-flow diagram for each process that is allowed. [10]

A certain Grand Unified Theory violates both $\mathbb{B}$ and $\mathbb{L}$, while conserving $\mathbb{Q}$. It contains a new boson X, with charge $\frac{1}{3}|e|$, and mass $m_{\mathrm{X}} \gg m_p$, where $m_p$ is the proton mass. The X boson interactions include the vertex corresponding to the process $\mathrm{X} \to u + d$ which has a dimensionless vertex factor $\lambda_{\mathrm{q}}$ and that corresponding to $\mathrm{X} \to e^+ + \bar{u}$ with a corresponding vertex factor $\lambda_\ell$.

(c) Assuming that all time orderings of these vertices are permitted, draw Feynman diagrams showing how the X boson could mediate the decay processes $p \to e^+ + \pi^0$ and $n \to e^+ + \pi^-$. [4]

(d) The expected proton lifetime resulting from such a process is of the form

$$\tau_p \approx A \frac{m_{\mathrm{X}}^4 \hbar}{\lambda_\ell^2 \lambda_{\mathrm{q}}^2 \, m_p^5 c^2} \, ,$$

where $A$ is a dimensionless constant. Justify the form of the dependence of $\tau_p$ on the indicated masses and couplings. [6]

**4.** (a) Define the term *differential cross section.* The leading order differential cross section for scattering from a localised potential $V(\mathbf{x})$ is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{m}{2\pi\hbar^2}\right)^2 \left| \int \mathrm{d}^3\mathbf{x}\; V(\mathbf{x})\, \mathrm{e}^{\mathrm{i}\,\mathbf{q}\cdot\mathbf{x}} \right|^2 .$$

Explain the meaning of the symbols $\mathbf{q}$ and $m$ in this equation. [4]

(b) Show for the potential

$$V_\mu(r) = V_0 \frac{\mathrm{e}^{-\mu r}}{r} ,$$

where $V_0$ and $\mu$ are constants, that

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{A}{(q^2+\mu^2)^2} ,$$

stating the value of the constant $A$.

Hence show that the differential cross section for Coulomb scattering is proportional to $\mathrm{cosec}^4(\Theta/2)$, where $\Theta$ is the scattering angle. [11]

(c) Draw a Feynman diagram for the process of elastic electron–electron scattering by neutral Higgs boson (H) exchange, representing the exchanged H boson by a dashed line. Show that the ratio $R_\mathrm{H}$ of the scattering matrix element for H exchange to that for photon exchange may be written

$$R_\mathrm{H} = \frac{\lambda_e^2}{4\pi\alpha} \frac{q^2}{q^2+\mu_\mathrm{H}^2} ,$$

where $\lambda_e$ is the coupling of the Higgs boson to the electron. [5]

Determine the maximum numerical value of $R_\mathrm{H}$ for elastic electron–electron collisions at centre-of-mass energy of 200 GeV. Comment briefly on the praticality of using this method to measure the force caused by Higgs boson exchange. [5]

[*You may assume that the Higgs boson has mass of 125* $\mathrm{GeV}/c^2$, *and that the coupling constant* $\lambda_e = 2.9 \times 10^{-6}$.]